CFMNet: A Change-Similarity Feature Mixing Network for Remote Sensing Change Detection

Cui Zhang, Liejun Wang, and Panpan Zheng

***Abstract*—In remote sensing image change detection (CD) tasks, establishing long-range dependencies between image tokens is crucial for comprehensive scene understanding and accurate localization of change areas. However, existing methods relying on distance-based similarity measures (e.g., Euclidean distance in ViG or dot product in ViT) face two key limitations: the mismatch between distance-driven similarity and semantic labels, and high computational complexity. To address these issues, we propose CFM network (CFMNet), a change-similarity feature mixing (CFM) network. Its core innovation is the CFM block, which achieves a paradigm shift in similarity measurement—from distance-driven to semantics-driven. Specifically, the CFM block employs a lightweight prediction head to assign a change confidence score to each token, representing its semantic identity. Tokens are then sorted and grouped based on their change confidence, ensuring that feature mixing (FM) optimization occurs only among tokens sharing the same semantic identity (changed or unchanged). This mechanism not only enables long-range feature interaction with linear time complexity but also resolves the mismatch between distance-driven similarity and semantic labels. It aligns the feature aggregation process with the ultimate task objective, avoiding redundant information transmission between semantically irrelevant tokens. To further enhance performance, CFMNet incorporates two additional components: a bidirectional feature enhancement (BFE) module, which facilitates complementary fusion of deep semantic information and shallow detailed features by constructing cross-layer information exchange paths, and dynamic KernelWarehouse (KW) convolution, which replaces static convolution as the encoder's fundamental operator by dynamically generating convolutional parameters based on input features, substantially boosting the model's feature representation capacity and generalization ability. Comprehensive experiments demonstrate that CFMNet achieves state-of-the-art performance on three CD datasets, outperforming other relevant methods.**

***Index Terms*—Change detection (CD), feature mixing (FM), remote sensing, semantics-driven learning.**

I. Introduction

REMOTE sensing image change detection (CD) technology [1], [2], [3] can generate change maps by analyzing surface images captured at the same location but at different times. This helps people understand the surface changes occurring at the interest location. The Earth's surface is constantly changing, including water and soil change, farmland change, woodland change, and building change [4], [5], [6], [7]. Therefore, CD technology holds significant research importance.

Early CD approaches based on deep learning primarily utilized convolutional neural networks (CNN) [8], [9], such as CDNet [1], FC-EF [2], STANet [10], SNUNet [11], and A2Net [12]. These methods leverage the local connectivity and weight-sharing properties of convolution operations, which excel at extracting fine-grained local details from images. However, the inherent locality of convolutions limits their ability to capture long-range features, thereby constraining their overall accuracy (OA) to some extent.

To address these limitations, many recent CD methods [3], [13], [14], [15] have incorporated Transformer [16], [17] or Mamba [18], [19] to enhance long-range feature extraction. In addition to these, graph convolutional network (GCN, e.g., ViG [20]) also provides a viable means for capturing long-range dependencies. We briefly summarize these three types of methods below

1) *GCN (Represented by ViG):* This approach measures token similarity by computing the Euclidean distance between their features, establishing connections between highly similar tokens for neighborhood aggregation. As it relies on distance-based similarity in the feature space to measure tokenwise affinity, it falls into the category of distance-driven similarity measurement methods. It suffers from two key drawbacks:
    a) *Mismatch Between Distance-Driven Similarity and Semantic Labels (Termed the Distance–Semantic Mismatch):* Tokens that are close in feature space (i.e., with small pairwise distances) may possess completely different semantic labels, while distant tokens might share the same label. As illustrated in Fig. 1, this mismatch causes the model to assign high similarity scores between the target token and semantically opposite ones. Consequently, redundant or even misleading information may be introduced during feature aggregation, ultimately leading to false positives or false negatives in detection.

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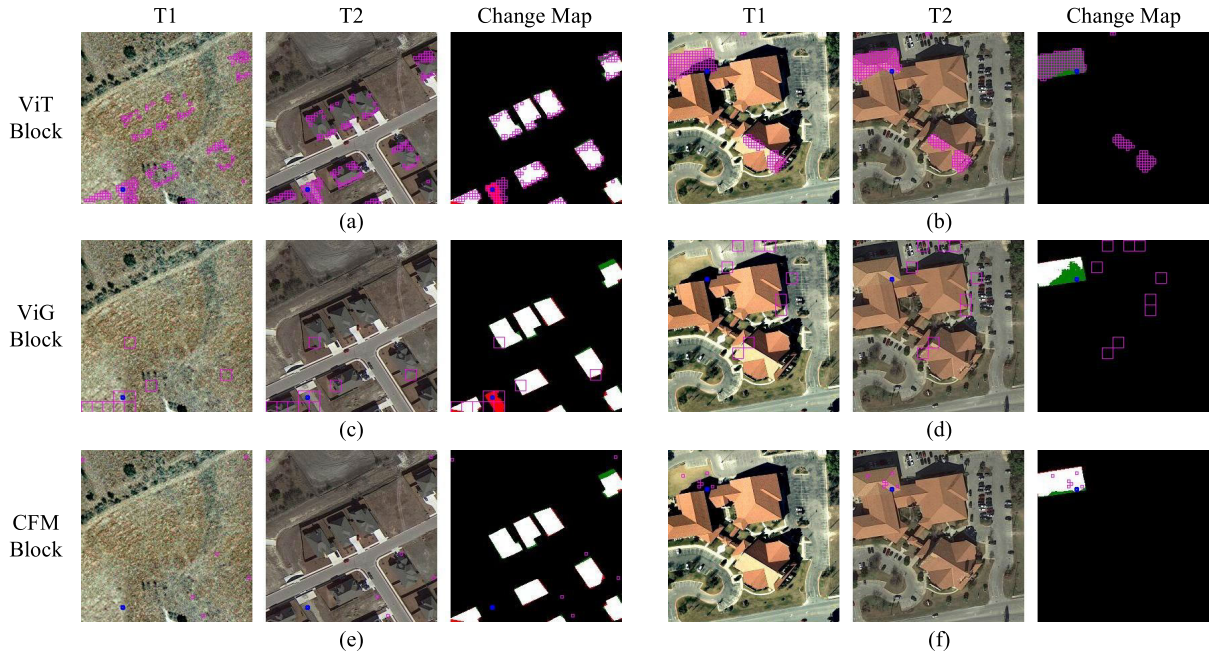


Fig. 1. Comparative visualization of similar token retrieval among three blocks. The target token is outlined in blue; its retrieved similar tokens are boxed in magenta. T1 and T2 are bitemporal images. Change maps highlight missed alarms in green and false alarms in red. (a) and (b) Visualize top-5% similar tokens retrieved by ViT block. (c) and (d) Visualize nine most similar tokens retrieved by ViG block (feature map downsampled before retrieval, so each token corresponds to a broader region). (e) and (f) Visualize seven most similar tokens retrieved by CFM block (including target).

b) *High Computational Complexity.* It requires computing pairwise distances between all tokens, leading to a computational complexity of $O(H^2W^2C)$.

2) *Transformer (Represented by ViT [17]):* This approach computes similarity weights between tokens using dot products (or scaled dot products) of their features, and updates each token via a weighted aggregation over all others. It also belongs to the category of distance-driven similarity measurement and suffers from the same two drawbacks: a high computational complexity of $O(H^2W^2C)$ and the distance–semantic mismatch. To our knowledge, this specific mismatch problem has not been explicitly identified or addressed in prior CD literature.
3) *Mamba (Represented by VMamba [19]):* This approach processes tokens sequentially, maintaining a hidden state to integrate historical context. The current token interacts with this state to capture long-range dependencies. While this design reduces computational complexity to linear, it introduces a bottleneck: the fixed-capacity hidden state may filter out or dilute long-range information, potentially limiting model performance.

To enhance the long-range feature extraction capability of CD methods, this article proposes a CFM network (CFMNet) for remote sensing CD. Its main innovation is the CFM block, which introduces a paradigm shift in token similarity measurement.

The CFM block shares a core idea with the ViG and ViT blocks: it computes token similarity and then performs feature mixing (FM) accordingly. However, it introduces a fundamental change in how similarity is measured. Instead of relying on distance-based similarity used in ViT and ViG (e.g., Euclidean distance or dot product), the CFM block adopts a semantics-driven similarity measurement. It first employs a lightweight prediction head to assign a change confidence score to each token. This score serves as a quantitative measure of the token's semantic identity (i.e., likelihood of being a change region). Tokens are then sorted and grouped based on these confidence scores, ensuring that FM occurs mainly among tokens with the same semantic identity (i.e., changed or unchanged). Its computational complexity is $O(HWC)$.

This mechanism resolves the distance–semantic mismatch (see Fig. 1). By restricting feature interaction to tokens of identical semantic identity, it prevents the aggregation of redundant or misleading information between unrelated tokens, thereby ensuring that the feature optimization process is aligned with the final task objective. Consequently, the CFM block achieves higher accuracy and stronger robustness with lower computational complexity than the ViT and ViG blocks.

To further improve the network's context-aware ability, we propose a bidirectional feature enhancement (BFE) module. The BFE module includes two feature streams. One stream flows from the higher layer to the lower layer, allowing each layer to incorporate the features of the higher layer. The other stream flows from the lower layer to the higher layer, enabling each layer to incorporate the features of the lower layer. The incorporation of BFE module allows the low-level features to capture the overall semantic information of the image, while the high-level features can perceive the detailed information in the image. This compensates for the respective limitations of each level's features and enhances the robustness of CFMNet.

In addition, this article utilizes KernelWarehouse (KW) convolution [21] to construct the encoder of CFMNet. This convolutional variant operates dynamically by adjusting its

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kernel parameters adaptively based on input features. Furthermore, to increase the number of kernel cells, KW convolution employs a strategy involving kernel partitioning and shared KWs. This design enhances both the generalization capacity and representational diversity of the network.

In summary, the primary contributions of this article are threefold

1) *Proposal of the CFM Block:* We propose the CFM block, which establishes a semantics-driven paradigm for long-range dependency modeling. Departing from conventional distance-driven similarity measures, CFM groups tokens based on their predicted semantic identity (change confidence), ensuring that feature aggregation occurs only among semantically consistent regions. This effectively resolves the distance–semantic mismatch while maintaining linear computational complexity.
2) *Proposal of the BFE Module:* We design a BFE module. It establishes top-down and bottom-up information pathways between deep and shallow layers, enabling complementary fusion of high-level semantics and low-level details. This enhances the network's ability to represent multiscale change contexts robustly.
3) *Superior Experimental Performance:* Comprehensive experiments on three public benchmarks (LEVIR-CD [10], CDD [22], and SYSU-CD [23]) demonstrate that our proposed CFMNet achieves state-of-the-art performance, outperforming other advanced CD methods both quantitatively and qualitatively.

II. Related Work

A. CNN-Based Methods

At present, CNN is one of the mainstream methods in CD field. CNN has intrinsic local characteristics and can effectively extract local features from images. To enhance CD performance, existing research primarily revolves around the following directions: optimization of network structure, incorporation of attention mechanisms, improvement of model efficiency, and addressing data-related challenges.

In terms of network structure, Caye Daudt et al. [2] propose an end-to-end trainable Siamese network framework, laying an important foundation for subsequent research. SNUNet [11] and DARNet [24] adopt densely connected structures, which facilitate the aggregation and reuse of multiscale features, effectively mitigating information loss during the decoding stage and improving gradient flow.

The incorporation of attention mechanisms has also become a common strategy in CD methods, aiming to enhance feature discriminability by focusing on spatiotemporal regions of semantic importance. STANet [10] designs a pyramid spatiotemporal attention module to handle changes of various sizes. DASNet [25] and SNUNet [11] integrate a dual attention module and an ensemble channel attention module, respectively, to recalibrate feature importance. DARNet [24] further proposes a hybrid attention module intended to capture long-range dependencies across spatial, temporal, and channel dimensions, partially compensating for the locality of convolutional operations.

In terms of efficiency improvement, USSFCNet [26] constructs an ultralightweight CD network based on multiscale decoupled convolution, while A2Net [12] builds upon the efficient MobileNetV2 backbone. These works demonstrate that careful structural design can significantly reduce computational complexity and parameter count while maintaining competitive performance.

Additionally, many studies focus on addressing common data-related challenges in remote sensing CD. To tackle the data imbalance problem, Tri-CD [27] employs a triplet loss function, while HANet [28] proposes a progressive foreground-balanced sampling strategy. To reduce reliance on costly pixel-level annotations, LGTC [29] introduces a semisupervised strategy that requires only a small number of labels, and SWCD [30] constructs a weakly supervised learning framework.

In summary, CNN-based methods have established an effective paradigm for CD. However, these methods are inherently limited in modeling long-range dependencies, which is crucial for understanding complex remote sensing scenes. This fundamental limitation serves as a primary motivation for researchers to explore new approaches with global modeling capabilities, such as Transformers and Mamba.

B. Transformer-Based Methods

Transformer [16], renowned for its success in natural language processing, has been introduced to the field of computer vision in recent years. Its core self-attention mechanism enables the modeling of long-range dependencies within images, effectively compensating for the limitations of CNN in understanding global context. With the advent of foundational vision Transformers such as ViT [17] and Swin Transformer [31], Transformer-based methods have emerged as a prominent line of research in CD. These methods generally fall into two types: hybrid Transformer–CNN architectures and Transformer backbone architectures.

In hybrid architectures, Transformer and CNN are used in conjunction to complement each other in extracting global and local features. BiT [3] first utilizes ResNet18 for feature extraction, followed by a Transformer encoder and decoder for global contextual modeling. TransUNetCD [32] and MSTDSNet-CD [33] adopt a similar strategy by integrating standard Transformer or Swin Transformer modules after convolutional networks to enhance the global correlation of features. ICIF-Net [34] designs a dual-branch architecture, where ResNet18 and PVTv2 [35] are employed in parallel to extract local and global features, respectively. AEGL-Net [36] uses MobileNetV2 as the backbone and incorporates a global–local feature fusion module based on self-attention mechanisms.

In Transformer backbone architectures, Transformer serves as the core feature extractor. ChangeFormer [13] proposes a hybrid network combining a Transformer backbone for feature extraction with a lightweight MLP decoder to recover change information. SwinSUNet [37] introduces a pure Transformer network for CD, which employs Swin Transformer blocks as its fundamental building units. HierFormer [38] designs

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a local—global-aware attention block to construct the backbone and reduces the computational overhead of self-attention through sparse attention mechanisms.

In summary, Transformer-based methods bring powerful global modeling to CD tasks, markedly enhancing the understanding of complex scenes. However, the self-attention mechanism suffers from quadratic time complexity with respect to input size, restricting its use in dense prediction tasks. Moreover, as a distance-driven similarity measurement, self-attention is prone to distance–semantic mismatch, potentially injecting irrelevant or even misleading information during feature aggregation. Therefore, reducing its computational cost and alleviating this mismatch remain open challenges for future research.

C. Mamba-Based Methods

Nowadays, numerous Mamba-based CD methods have been proposed [14], [15], [39]. Similar to Transformer, Mamba [18] excels in global feature extraction. ChangeMamba [14] is a CD network entirely based on the visual state space block. RS-Mamba [39] constructs a CD network using the omnidirectional state space block, enabling the scanning of image information from eight directions. CDMamba [15] is primarily built from the ConvMamba block, which integrates both convolution and Mamba operations.

In summary, these Mamba-based CD methods have achieved remarkable results. However, Mamba's global feature extraction ability largely depends on the extracted hidden states, and the capacity of these hidden states restricts the network's performance to some extent.

D. Other Methods

Beyond the mainstream methods described above, several CD techniques exploit alternative paradigms. For example, CGSL [40] is a GCN-based CD method that converts heterogeneous images into superpixel graphs and employs a Siamese graph encoder to extract modality-invariant graph representations for comparison. AEKAN [41] is a multimodal CD method based on Kolmogorov–Arnold network [42]. It employs a Siamese encoder to extract modality-invariant commonality features and dual decoders to reconstruct the original bitemporal images, effectively enabling unsupervised comparison of heterogeneous data.

These methods introduce novel paradigms and perspectives to CD, broadening the technical landscape. However, the fundamental challenge of efficiently establishing long-range dependencies remains an open problem for these alternative approaches.

III. Methodology

In this article, we propose a CFMNet for remote sensing CD, whose architecture is illustrated in Fig. 2. It primarily consists of four components: 1) an encoder employing dynamic KW convolution [21]; 2) a BFE module designed to strengthen multiscale contextual awareness; 3) a dual–temporal feature fusion (DTFF) module; and 4) a decoder incorporating a CFM block to achieve semantics-driven and efficient long-range interaction. The details of each component will be elaborated in Sections III-A–III-D.

A. Encoder Based on KW Convolution

The encoder adopts MobileNetV2 [43] as its foundational architecture. To enhance feature representation capacity and model generalization, we replace the standard static convolutions with dynamic KW convolutions while preserving the overall structural topology.

1) KW Convolution: DY-Conv [44] enhances convolutional operations by transforming static kernels into dynamic ones. Specifically, it generates the dynamic kernel W as a linear combination of n static kernels

$$W = \alpha_1 W_1 + \cdots + \alpha_n W_n \tag{1}$$

where W_i denotes the i_{th} static kernel and α_i represents the corresponding combination coefficient. These coefficients are dynamically conditioned on the input features, typically generated via an attention mechanism. While increasing n can improve representational capacity, it also leads to a linear growth in the number of parameters. As a result, n is usually kept small (e.g., 4 or 8) in practice.

KW convolution is different from DY-Conv. To increase the number of static kernels, it utilizes kernel partition and shared kernel cell warehouses techniques. As shown in Fig. 3, the convolutional kernel W is first partitioned into m equally sized local kernels

$$\begin{aligned} W = w_1 \cup \cdots \cup w_m \quad \forall i,\ j \in \{1, \ldots, m\} \\ i \neq j,\ w_i \cap w_j = \emptyset \end{aligned} \tag{2}$$

where w_i is referred to as the local kernel, and m represents the number of local kernels. KW designs a kernel cell warehouse, containing n kernel cells e_i, each with the same size as w_i. Each w_i is obtained by weighting all kernel cells and is calculated as follows:

$$w_i = \alpha_{i1} e_1 + \cdots + \alpha_{in} e_n, i \in \{1, \ldots, m\} \tag{3}$$

the weight of the i_{th} local kernel w_i with respect to the j_{th} kernel cell e_j, calculated by an attention operation, is denoted as α_{ij}, where $j \in \{1, 2, \ldots, n\}$. By kernel partition, the kernel cell number n can increase significantly.

Moreover, kernel cell warehouses are shared across all convolutional layers within the same stage of the network, further significantly increasing the number of kernel cells accessible to each layer without extra parameters.

2) Hierarchical Structure of the Encoder: Following the paradigm of most previous CD methods [2], [37], CFMNet employs a Siamese encoder with shared weights between the dual–temporal branches. Our encoder is built upon a KW MobileNetV2 backbone. The core building block of this backbone is the KW MobileNetV2 block, whose detailed architecture is shown in Fig. 4. As illustrated, this block retains the complete inverted residual design of the standard MobileNetV2 block but replaces all its static convolutions with the dynamic KW convolutions described in Section III-A1. The encoder consists of four hierarchical stages, each primarily composed of these enhanced blocks.

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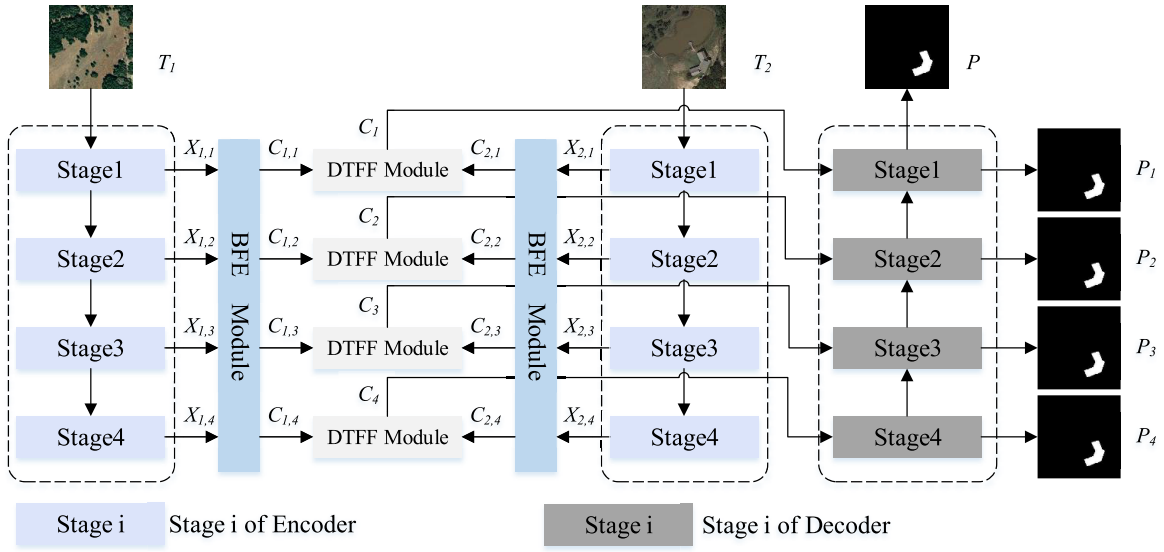


Fig. 2. Framework overview of CFMNet. The Siamese encoder extracts multiscale features $\{X_{1,i}\}_{i=1}^{4}$ and $\{X_{2,i}\}_{i=1}^{4}$ from input images T_1 and T_2, respectively. The BFE module enhances these features via bidirectional streams to produce $\{C_{1,i}\}_{i=1}^{4}$ and $\{C_{2,i}\}_{i=1}^{4}$. The DTFF module then fuses each feature pair $(C_{1,i}, C_{2,i})$ into C_i, yielding $\{C_i\}_{i=1}^{4}$. Finally, the CFM-based decoder integrates $\{C_i\}_{i=1}^{4}$ to generate the final change map P.

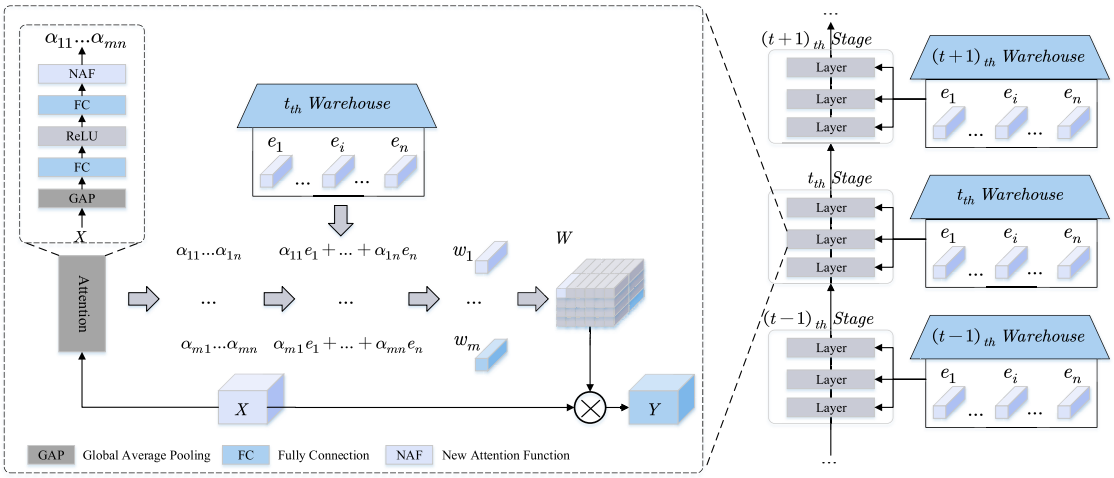


Fig. 3. Illustration of the KW convolution. The convolution kernel W is partitioned into multiple local kernels $\{w_i\}_{i=1}^{m}$. Each local kernel is dynamically generated as a weighted combination of kernel cells $\{e_j\}_{j=1}^{n}$ from shared warehouses. The combination weights are conditioned on the input feature X.

The encoder input consists of a pair of dual–temporal images T_1 and T_2. For each input image, the encoder will generate a set of multiscale features $X_{t,1}, X_{t,2}, X_{t,3}$, and $X_{t,4}$, where $t \in \{1, 2\}$. The spatial size of these four features gradually decreases, and the contained information gradually transitions from detailed to semantic.

B. BFE Module

After the image is processed by the encoder, the resulting multiscale features $X_{t,1}, X_{t,2}, X_{t,3}$, and $X_{t,4}$ are passed through a BFE module. The module enhances feature representation at each level by incorporating both fine-grained details from lower layers and high-level semantics from deeper layers. This bidirectional integration of multiscale features endows the network with enhanced contextual awareness and a more comprehensive understanding.

The BFE module begins by processing multiscale features through two separate convolutional layers, as illustrated in Fig. 5(a). The specific process is as follows:

$$f_{t,i} = Conv1 \times 1\left(X_{t,i}\right) \tag{4}$$

$$s_{t,i} = Conv3 \times 3\left(X_{t,i}\right) \tag{5}$$

where $i \in \{1, 2, 3, 4\}$ represents the stage index. $Conv1 \times 1$ represents a convolutional layer with a kernel size of 1×1, while $Conv3 \times 3$ represents a convolutional layer with a kernel size of 3×3. $f_{t,i}$ and $s_{t,i}$ represent the outputs of these two convolutional layers, respectively.

Then, features $s_{t,1}, s_{t,2}, s_{t,3}$, and $s_{t,4}$ are processed through upward and downward streams, respectively, as illustrated in Fig. 5(c). Each stream brings new information to every stage feature, except for the starting feature. The upward stream conveys comprehensive semantic information to the low-level

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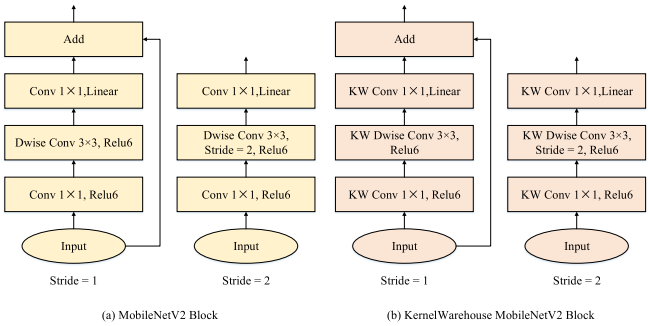


Fig. 4. Architectural diagrams of the (a) standard MobileNetV2 block and (b) its KW-enhanced counterpart. Both blocks share identical architectural topology, including the 1×1 pointwise expansion convolution, 3×3 depthwise convolution, and 1×1 pointwise projection convolution. The sole distinction is that all convolutions in (b) employ dynamic KW convolution, whereas those in (a) use standard static convolution.

features, while the downward stream conveys detailed information to the high-level features. The feature flow in the upward stream is computed as follows:

$$p_{t,i}^{U} = s_{t,i},\ i = 4 \tag{6}$$

$$p_{t,i}^{U} = Conv3\times3\left(Concat\left(s_{t,i}, US\left(p_{t,i+1}^{U}\right)\right)\right),\ i \neq 4 \tag{7}$$

where US denotes the upsampling operation (implemented using bilinear interpolation), and $Concat$ refers to concatenation along the channel dimension. The feature flow in the downward stream is computed as follows:

$$p_{t,i}^{D} = s_{t,i},\ i = 1 \tag{8}$$

$$p_{t,i}^{D} = Conv3\times3\left(Concat\left(s_{t,i}, DS\left(p_{t,i-1}^{D}\right)\right)\right),\ i \neq 1 \tag{9}$$

where DS denotes the downsampling operation (implemented using max-pooling). Following the bidirectional processing, the BFE module combines the resulting upward and downward features $p_{t,i}^{U}$ and $p_{t,i}^{D}$ through a merging operation. This process can be expressed as follows:

$$p_{t,i} = Conv3\times3\left(Concat\left(p_{t,i}^{D}, p_{t,i}^{U}\right)\right). \tag{10}$$

Finally, the BFE module employs a residual connection to integrate $f_{t,i}$ with $p_{t,i}$ via elementwise addition, as depicted in Fig. 5(d). This process is defined as follows:

$$C_{t,i} = f_{t,i} + p_{t,i}. \tag{11}$$

where $C_{t,1}, C_{t,2}, C_{t,3}$, and $C_{t,4}$ are the final output results of the BFE module.

C. DTFF Module

After processing through the BFE module, two sets of multiscale features are generated for the bitemporal image pair, denoted as $\{C_{1,1}, C_{1,2}, C_{1,3}, C_{1,4}\}$ and $\{C_{2,1}, C_{2,2}, C_{2,3}, C_{2,4}\}$. To prepare these features for the subsequent decoder, CFMNet merges them through a DTFF module.

The fusion process at each scale i is formulated as follows:

$$D_i = Conv3\times3\left(|C_{1,i} - C_{2,i}|\right) \tag{12}$$

$$C_i = Conv3\times3\left(D_i\right) + D_i \tag{13}$$

where $|\cdot|$ denotes the elementwise absolute value operation. The output $\{C_1, C_2, C_3, C_4\}$ is the set of fused multiscale features passed to the decoder.

D. Decoder Based on CFM Block

Similar to the encoder, the decoder of CFMNet follows a four-level hierarchical structure, as shown in Fig. 6(a). The first three stages of the decoder each contain three components: a multifeature fusion block (MFF block), a convolution block (Conv block), and a CFM block. The fourth stage consists of a Conv block followed by a CFM block. In the decoder, the Conv block extracts local detailed features, while the CFM block models global semantic dependencies. These two components operate synergistically, integrating fine-grained local information with high-level contextual understanding to form a comprehensive and discriminative feature representation.

1) MFF Block: In the first three stages of the decoder, the input to the i_{th} stage consists of two features: one is the feature C_i generated by the corresponding stage in the DTFF module, and the other is the upsampled version of the output from the $(i+1)_{th}$ decoder stage, denoted as dm_{i+1}^{U}. These two features are first fused using the MFF block. The purpose of introducing feature C_i is to reintroduce fine-grained details that were lost during previous downsampling operations.

The detailed operation of the MFF block is depicted in Fig. 6(b). The C_i is first processed by a 3×3 convolution. dm_{i+1}^{U} is processed by a 3×3 convolution and a 1×1 convolution, respectively,

$$df1_i = Conv3\times3\left(C_i\right) \tag{14}$$

$$df1_i^{U} = Conv3\times3\left(dm_{i+1}^{U}\right) \tag{15}$$

$$df2_i^{U} = Conv1\times1\left(dm_{i+1}^{U}\right). \tag{16}$$

After the convolution process, the resulting features $df1_i$ and $df1_i^{U}$ are merged together using a channel concatenation operation. Subsequently, a 3×3 convolution is applied to reduce the number of feature channels to match that of $df1_i$

$$df2_i = Conv3\times3\left(Concat\left(df1_i, df1_i^{U}\right)\right). \tag{17}$$

The merged features are added to $df2_i^{U}$ via a residual connection. The added feature df_i is the output of the MFF block

$$df_i = df2_i + df2_i^{U}. \tag{18}$$

2) Conv Block: The decoder's Conv block mainly comprises two convolutional layers and a residual connection, as illustrated in Fig. 6(c). The specific implementation formula for the Conv block is as follows:

$$dc1_i = Conv3\times3\left(df_i\right) \tag{19}$$

$$dc2_i = Conv3\times3\left(dc1_i\right) \tag{20}$$

$$dc_i = dc1_i + dc2_i. \tag{21}$$

The input of the Conv block df_i ($df_4 = C_4$) will pass through two consecutive convolutional layers, resulting in two features $dc1_i$ and $dc2_i$. These two features are then added together via a residual connection to generate dc_i, which represents the output of the Conv block.

3) CFM Block: To complement the local receptive field of the Conv block, we introduce the CFM block for modeling long-range dependencies. Unlike the ViT and ViG blocks, which measure token similarity based on feature distance (e.g., dot product or Euclidean distance), the CFM block

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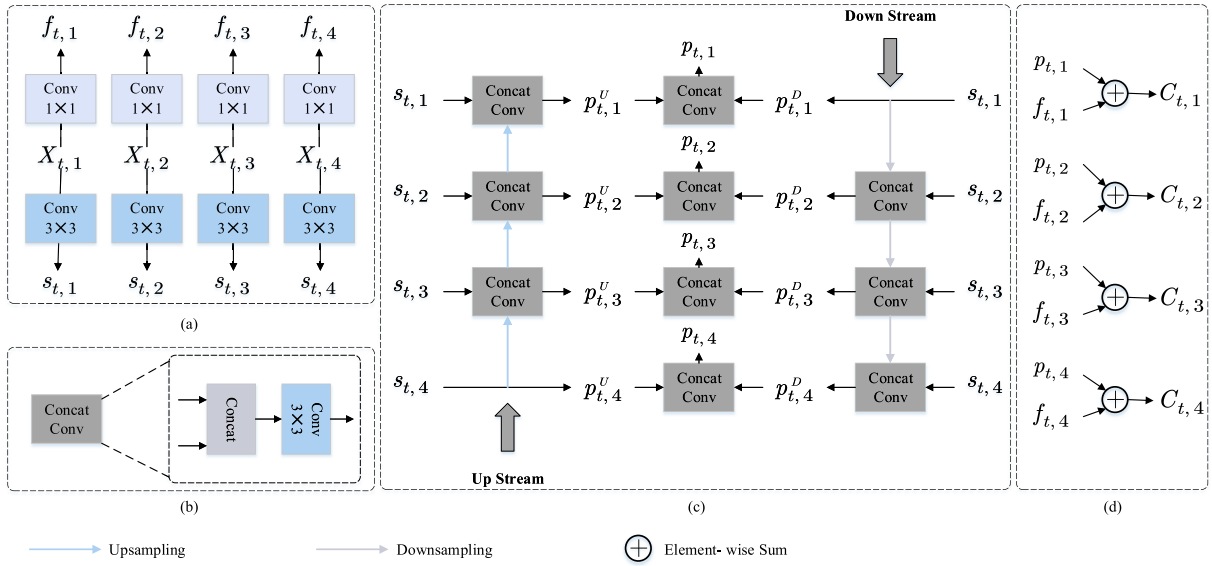


Fig. 5. Structure of the BFE module. (a), (c), and (d) Three main steps of the BFE module. (b) Implementation flow of the ConcatConv operation.

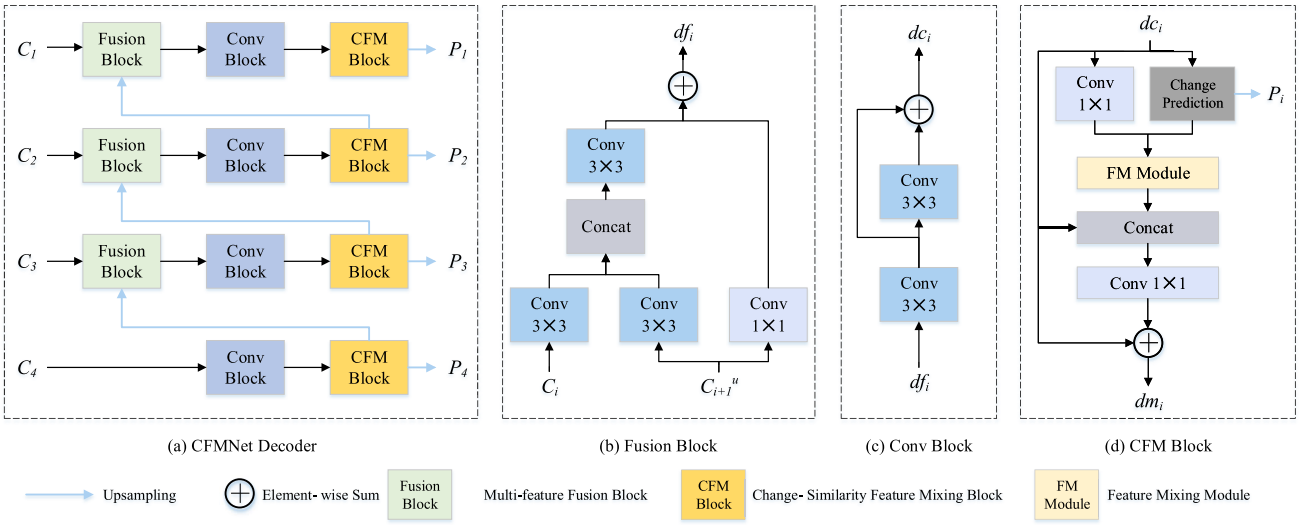


Fig. 6. Structure of the CFMNet decoder. (a) Overall structure of the decoder. (b) MFF block. (c) Conv block. and (d) CFM block.

adopts a semantics-driven similarity measurement: tokens are considered similar only if they have similar predicted change confidence scores. Following this measurement, the CFM block enables each token to interact with its k-most semantically similar tokens. Consequently, every long-range interaction is explicitly aligned with the CD objective while maintaining near-linear complexity.

The CFM block executes four consecutive steps [Fig. 6(d)].

Step 1 (Semantic Guidance Generation): A lightweight change-prediction head produces a pixelwise change confidence map

$$Scores_i = SoftMax\left(Conv1 \times 1\left(dc_i\right)\right). \tag{22}$$

Each value in $Scores_i$ represents the predicted probability that its associated token belongs to the change class. This prediction is directly supervised by comparing an upsampled version P_i with the ground truth map via an auxiliary loss. Consequently, $Scores_i$ is not merely an intermediate feature but a reliable, task-level semantic signal, providing guidance for subsequent operations.

Step 2 (Feature Projection): The CFM block utilizes a 1×1 convolution to process dc_i and obtain $dm1_i$

$$dm1_i = Conv1 \times 1\left(dc_i\right). \tag{23}$$

This projection adapts the feature representation for the subsequent semantics-guided mixing operation.

Step 3 [Semantics-Guided FM (Core)]: The CFM block employs the FM module to update $dm1_i$ via the semantic signal $Scores_i$, resulting in the feature $dm2_i$. The formula is as follows:

$$dm2_i = FM\left(dm1_i, Scores_i\right). \tag{24}$$

The FM module refines features by performing three sequential operations, as detailed in Fig. 7

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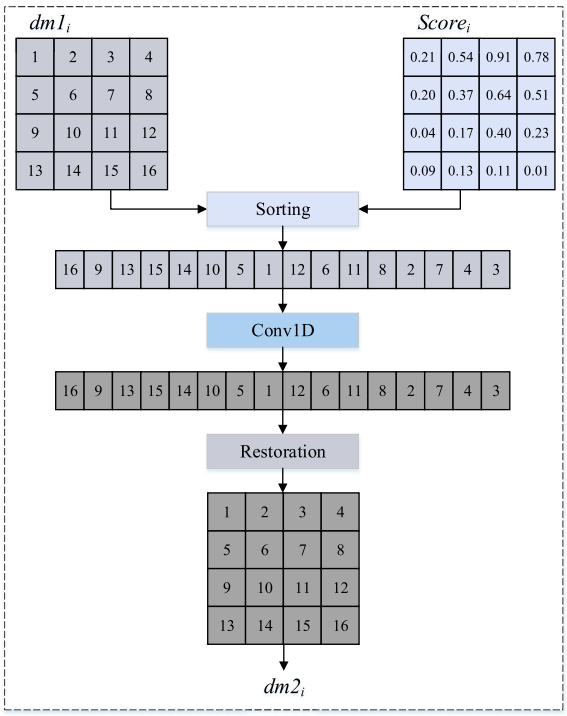


Fig. 7. Structure of the FM module, the core component of the CFM block.

1) *Semantic-Guided Sorting:* All tokens in $dm1_i$ are rearranged into a 1-D sequence z in ascending order of their change confidence scores (from $Scores_i$). Consequently, tokens with low change probability are automatically distributed at one end of z, while those with high probability are distributed at the other end [visualized in Fig. 7]. This sorting creates semantically homogeneous neighborhoods within the sequence.
2) *Semantic-Driven Neighborhood Interaction (1-D Convolution):* A kernel of size k slides over z, allowing each token to interact only with its k semantically nearest neighbors (including itself). Because the neighborhood is determined by semantic confidence rather than feature space distance, the convolution ensures that features are mixed primarily within the same semantic category (changed versus unchanged).
3) *Spatial Restoration:* The refined sequence is mapped back to the original spatial layout, yielding $dm2_i$. The restoration is the inverse permutation of semantic-guided sorting.

Step 4 (Residual Fusion): The original feature dc_i and the refined feature $dm2_i$ are first concatenated, then fused by a 1×1 convolution, and finally integrated via a residual connection to produce the output dm_i

$$dm3_i = Conv1 \times 1\,(Concat\,(dc_i, dm2_i)) \tag{25}$$

$$dm_i = dc_i + dm3_i. \tag{26}$$

CFM block has two advantages:

a) Near-linear computational complexity: Compared to ViT and ViG blocks, which require computing all pairwise token similarities with $O(H^2W^2C)$ complexity, the CFM block reduces this to $O(HW\log(HW))$ for sorting plus $O(HWC)$ for FM. In practical applications, the $O(HW\log(HW))$ term is typically smaller than the $O(H \times WC)$ term, resulting in an approximately linear computational complexity and significantly higher computational efficiency.

b) Semantic-Driven Interaction: By employing a loss-supervised change prediction head, the CFM block aligns token mixing with the semantic goal of CD. Token interaction occurs only between tokens belonging to the same semantic category, which enhances representational relevance and reduces noise, leading to more discriminative and robust features.

IV. Experiments

A. Datasets

In this article, three public datasets are used in the experiments, they are LEVIR-CD [10], CDD [22], and SYSU-CD [23]. In this section, we will briefly introduce each of them.

The LEVIR-CD dataset has 637 sets of images, which are derived from Google Earth. These images, which focus on building changes, were taken in Texas, USA from 2002 to 2018. They have a size of 1024×1024 pixels and a spatial resolution of 0.5 m/pixel. In the experiment, they are divided into three subsets: training, test and validation. The number of images in these subsets is 445, 128, and 64, respectively.

CDD dataset contains seven sets of 4725×2200 images and four sets of 1900×1000 images. Their spatial resolution is 0.03–1 m/pixel. To facilitate the experiment, the images were cut into 16 000 256×256 image patches. In practice, these image patches are split into training, test, and validation subsets with the number of patches being 10 000, 3000, and 3000, respectively. This dataset labels multiple types of changes, including building changes, car changes, and road changes.

SYSU-CD dataset has 20 000 sets of 256×256 aerial images. These images were taken from 2007 to 2014 in Hong Kong, China. Their spatial resolution is 0.5 m/pixel. In the specific application, they are split into training, test, and validation subsets with the image number being 12 000, 4000, and 4000, respectively. This dataset labels multiple types of changes, including building changes, vegetation changes, road changes, and sea construction changes.

B. Evaluation Criteria

In our experiments, we utilize five evaluation metrics to evaluate the CD method. They are precision, recall, OA, $F1$-score, and intersection over union (IoU). Among them, the latter two are more important metrics. Before calculating these metrics, we need to first calculate true positive number (TP), true negative number (TN), false positive number (FP), and false negative number (FN). Then, compute them according to the following formulas:

$$precision = \frac{TP}{TP + FP} \tag{27}$$

$$recall = \frac{TP}{TP + FN} \tag{28}$$

$$F1 = \frac{2}{precision^{-1} + recall^{-1}} \tag{29}$$

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TABLE I
COMPARISON RESULTS ON LEVIR-CD TEST SET

Method	Pre. (%)	Rec. (%)	F1 (%)	OA (%)	IoU(%)
SNUNet [11]	91.47	89.13	90.28	99.02	82.29
BIT [3]	92.34	87.35	89.78	99.00	81.45
ChangeFormer [13]	91.84	87.45	89.59	98.98	81.15
USSFCNet [26]	88.81	90.97	89.88	98.98	81.61
A2Net [12]	92.29	90.76	91.52	99.14	84.36
SCD-SAM [45]	92.28	89.72	90.98	99.09	83.45
ChangeMamba [14]	**95.11**	86.01	90.33	99.06	82.36
CDMamba [15]	91.41	90.05	90.72	99.06	83.02
HierFormer [38]	92.63	89.19	90.88	99.09	83.28
AEGL-Net [36]	93.02	89.88	91.43	99.14	84.21
CFMNet	92.45	**91.26**	**91.85**	**99.18**	**84.93**

TABLE II
COMPARISON RESULTS ON CDD TEST SET

Method	Pre. (%)	Rec. (%)	F1 (%)	OA (%)	IoU(%)
SNUNet [11]	95.83	93.49	94.65	98.63	89.84
BIT [3]	96.68	93.92	95.28	98.80	90.98
ChangeFormer [13]	95.44	94.01	94.72	98.65	89.97
USSFCNet [26]	94.64	96.08	95.36	98.81	91.12
A2Net [12]	98.10	97.96	98.03	99.51	96.14
SCD-SAM [45]	96.50	96.22	96.36	99.10	92.97
ChangeMamba [14]	97.16	97.71	97.43	99.39	94.99
CDMamba [15]	96.71	94.90	95.79	98.97	91.92
HierFormer [38]	97.25	97.23	97.24	99.32	94.63
AEGL-Net [36]	97.95	97.82	97.88	99.48	95.85
CFMNet	**98.50**	**98.32**	**98.41**	**99.61**	**96.87**

$$OA = \frac{TP + TN}{TP + TN + FN + FP} \tag{30}$$

$$IoU = \frac{TP}{TP + FN + FP}. \tag{31}$$

C. Experimental Settings

In the experiments, our models are trained on NVIDIA TITAN RTX GPU using PyTorch. During training, we used polylearning scheme to adjust the learning rate (LR). The LR is calculated as follows:

$$lr = \left(1 - \frac{i}{iterations}\right)^{power} \times lr_{initial} \tag{32}$$

where i denotes the current iteration index; *iterations* represents the total number of iterations, which is set to 40 000 (LEVIR-CD and SYSU-CD) or 160 000 (CDD) in our experiments; $lr_{initial}$ denotes the initial LR, which is set to 0.0005; and *power* is set to 0.9.

D. Comparative Experiments

We compare CFMNet with other representative CD methods. SNUNet [11], USSFCNet [26], and A2Net [12] are three CNN-based CD methods. BiT [3], ChangeFormer [13], HierFormer [38], and AEGL-Net [36] are four Transformer-based CD methods. SCD-SAM [45] is a SAM-based CD method. ChangeMamba [14] and CDMamba [15] are both Mamba-based CD methods.

1) Comparative Experiments on LEVIR-CD Dataset: Table I presents the quantitative results on the LEVIR-CD test set. CFMNet achieves the best overall performance, leading to $F1$, OA, *recall*, and IoU. A key observation is that while ChangeMamba achieves the highest *precision* (95.11%), its *recall* is significantly lower (86.01%). This indicates a tendency toward overly conservative predictions, likely missing subtle changes. In contrast, CFMNet achieves a superior balance, attaining both high *precision* (92.45%) and the highest *recall* (91.26%). Compared with the suboptimal method A2Net, CFMNet improves $F1$ and IoU by 0.33% and 0.57%, respectively.

The qualitative results on the LEVIR-CD dataset are visualized in Fig. 8, which displays the change maps generated by various CD methods for four samples. In Fig. 8(a), several building foundations are visible in the T1 image, while some completed buildings are present in the T2 image. Due to the similarity between the building foundation and the completed building, many methods fail to accurately identify the changed areas. Fig. 8(b) and (d) includes changed regions with slightly more complex appearances, which can hinder some methods from recognizing them correctly. Fig. 8(c) contains multiple changed regions with small relative areas, and most methods struggle to accurately locate their boundaries. Overall, it is evident that the change map generated by CFMNet is more accurate and closely aligns with the actual change map.

Combining quantitative and qualitative results, ChangeMamba's quantitative metrics show a low $F1$ with high *precision* but low *recall*, and its visualization results exhibit many missed detections and few false positives, which is consistent between the two. This phenomenon is primarily attributed to the characteristics of the LEVIR-CD dataset, where change targets are predominantly buildings that are regular in shape and small in size. Consequently, in addition to global context understanding, the precise extraction of local details becomes the determining factor for detection performance. ChangeMamba employs a four-direction scanning selective state space model (SSM). While this approach excels at global context modeling, it has limitations in local feature extraction. Consequently, its performance is constrained when handling the regular, small-scale change targets prevalent in the LEVIR-CD dataset. CFMNet achieves superior performance in both quantitative and qualitative evaluations. This is attributed not only to the global context understanding of the CFM block, but also to the local modeling foundation established by KW convolutions, as well as the precise capture of boundary details enabled by the bidirectional multiscale enhancement of the BFE module.

2) Comparative Experiments on CDD Dataset: On the CDD dataset, CFMNet achieves the highest values for all metrics. The quantitative analysis results of all methods are shown in Table II. From the data in the table, it can be seen that the $F1$ and IoU of CFMNet are at least 0.38% and 0.73% higher than those of other methods, respectively, which achieves high CD accuracy.

The qualitative results on the CDD dataset are visualized in Fig. 9, which displays the change maps generated by various CD methods for four samples. The change regions in Fig. 9(a) and (c) are caused by cars and buildings, the change regions in Fig. 9(b) are mainly caused by cars, and the change regions in Fig. 9(d) are caused by the road. For these change regions with

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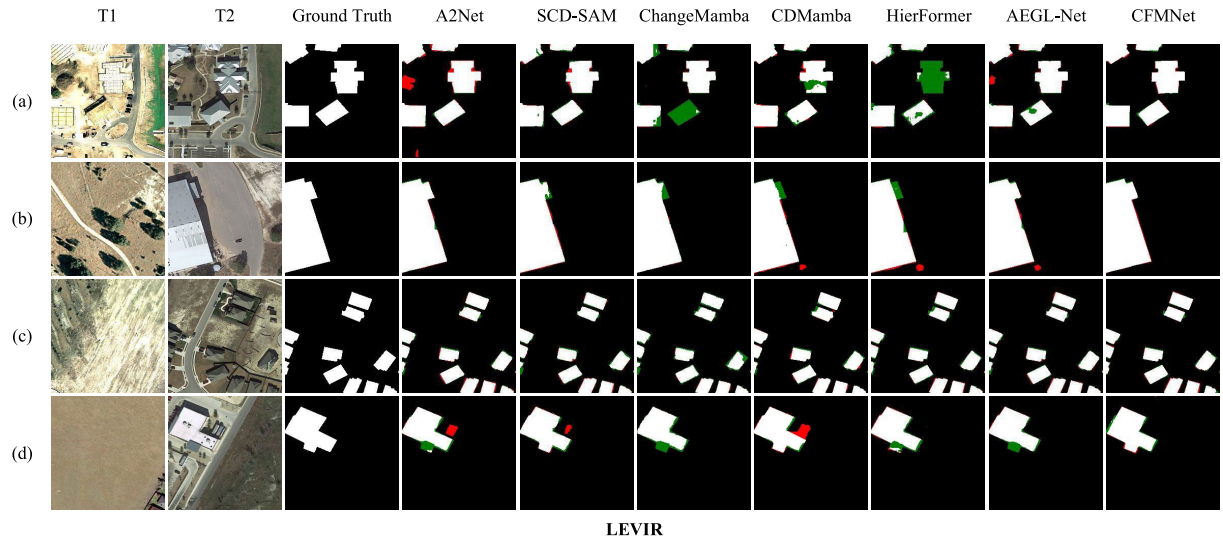


Fig. 8. Visualization results of several CD methods on the LEVIR-CD test set. (a)–(d) Present four sample sets selected from the test set, along with the corresponding change maps generated by the comparative methods. In the change maps, white, black, red, and green denote true positives, true negatives, false positives, and false negatives, respectively. The less red and green in the generated change map, the better the method is.

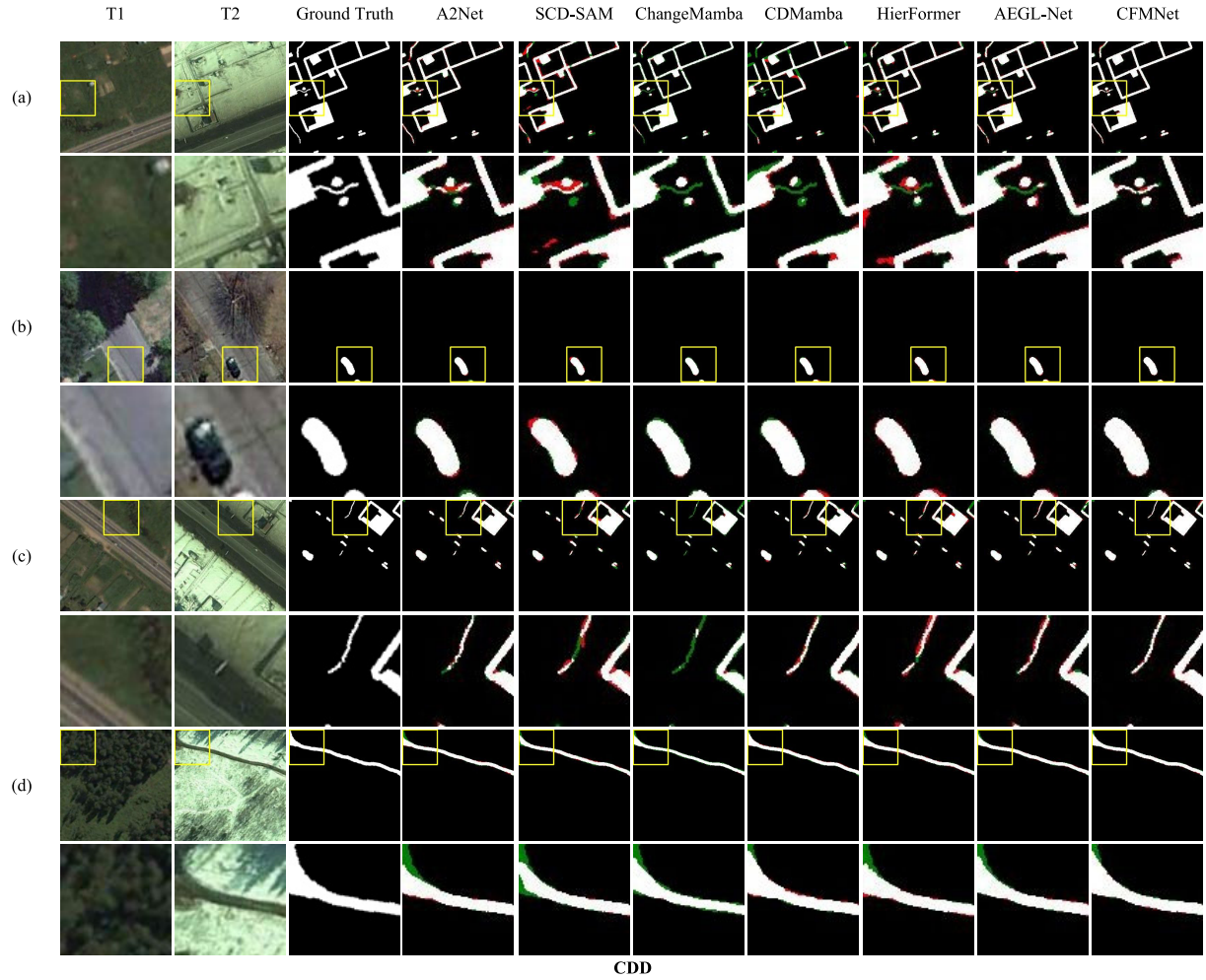


Fig. 9. Visualization results of several CD methods on the CDD test set. For clearer comparison, zoomed-in views (yellow boxes) highlight boundary-detail differences. (a)–(d) Present four sample sets selected from the test set, along with the corresponding change maps generated by the comparative methods. In the change maps, white, black, red, and green denote true positives, true negatives, false positives, and false negatives, respectively. The less red and green in the generated change map, the better the method is.

different sizes, all experimental methods generated relatively accurate change maps. However, the differences among the methods are primarily reflected in the edge refinement of the change regions. Compared with other CD methods, the change

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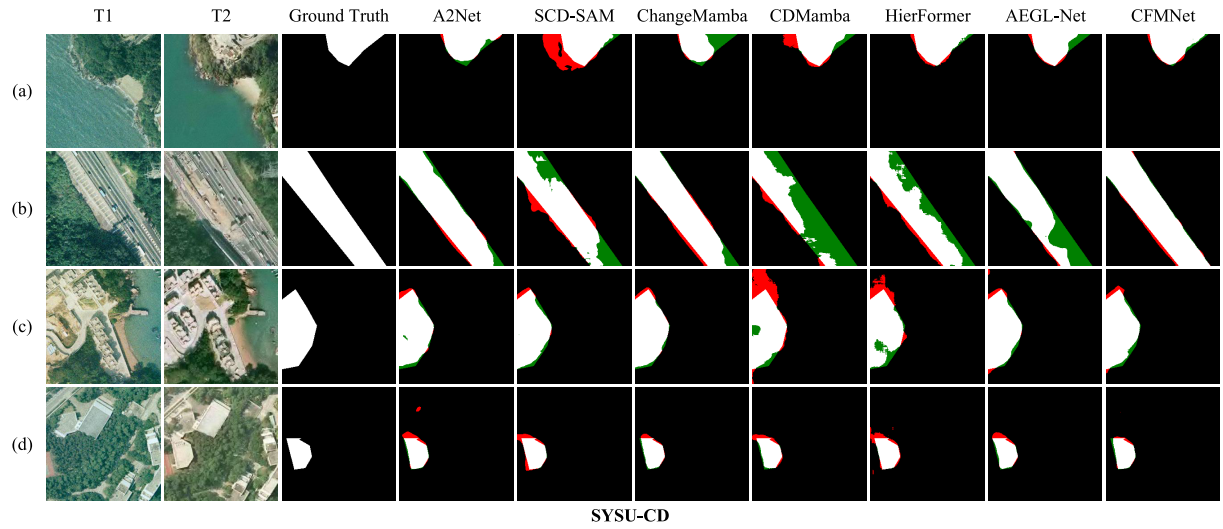


Fig. 10. Visualization results of several CD methods on the SYSU-CD test set. (a)–(d) Present four sample sets selected from the test set, along with the corresponding change maps generated by the comparative methods. In the change maps, white, black, red, and green denote true positives, true negatives, false positives, and false negatives, respectively. The less red and green in the generated change map, the better the method is.

maps generated by CFMNet are more precise at the edges and nearly identical to the ground truth, significantly reducing false positives and false negatives along boundaries.

Synthesizing quantitative and qualitative results, CDMamba yields a relatively low $F1$ of 95.79%. It performs poorly on the complex multiobject scene in Fig. 9(a), the partial vehicle regions at the bottom of Fig. 9(b), and the blurred buildings in the upper right of Fig. 9(c). These scenarios share common characteristics: ambiguous targets and difficulty in distinguishing foreground from background, which necessitate strong capabilities in high-level semantic understanding and context reasoning. Although CDMamba employs the ConvMamba module to fuse convolutional local perception with Mamba global modeling, its convolutional branch has a limited receptive field, and the Mamba branch is constrained by the 1-D nature of SSM, thereby restricting 2-D spatial information modeling. Consequently, it exhibits limited capability in handling such complex scenarios. Notably, CDMamba still achieves competent detection results in specific scenarios, such as the whole car in Fig. 9(b) and the road in Fig. 9(d). In these cases, where change regions are regular in shape and have relatively clear contours, the convolutional and Mamba branches can leverage their respective strengths to achieve satisfactory results. CFMNet delivers the best performance in both quantitative and qualitative analyses. By leveraging the BFE module to bolster fine-grained detail preservation and high-level semantic understanding, and by employing the CFM block to facilitate mutual enhancement between semantically related tokens, CFMNet significantly improves contextual understanding and effectively mitigates semantic confusion.

3) Comparative Experiments on SYSU-CD Dataset: CFMNet also achieves the best overall performance on the SYSU-CD dataset. The evaluation results of the various experimental methods are presented in Table III. As on the LEVIR-CD dataset, ChangeMamba continues to exhibit the characteristic of high *precision* but low *recall* (*precision*: 87.20%, *recall* : 78.79%). In contrast, CFMNet achieves the highest *recall* (83.00%) while maintaining high *precision* (85.59%). This demonstrates its superior capability in capturing a wider range of true changes without compromising accuracy. Compared with A2Net which achieves suboptimal results, CFMNet improves $F1$ by 1.18% and IoU by 1.73%.

TABLE III
COMPARISON RESULTS ON SYSU-CD TEST SET

Method	Pre. (%)	Rec. (%)	F1 (%)	OA (%)	IoU(%)
SNUNet [11]	80.06	79.25	79.65	90.45	66.19
BIT [3]	80.32	76.17	78.19	89.98	64.19
ChangeFormer [13]	82.21	75.63	78.78	90.39	64.99
USSFCNet [26]	77.84	79.69	78.75	90.09	64.95
A2Net [12]	86.36	80.09	83.10	92.32	71.09
SCD-SAM [45]	84.95	79.83	82.31	91.91	69.94
ChangeMamba [14]	**87.20**	78.79	82.78	92.27	70.63
CDMamba [15]	81.35	77.97	79.63	90.59	66.15
HierFormer [38]	82.30	75.92	78.98	90.47	65.27
AEGL-Net [36]	86.39	78.33	82.16	91.98	69.73
CFMNet	85.59	**83.00**	**84.28**	**92.70**	**72.82**

The qualitative results on the SYSU-CD dataset are visualized in Fig. 10, which displays the change maps generated by various CD methods for four samples. The changes in Fig. 10(a) are mainly caused by vegetation, those in Fig. 10(b) by roads, those in Fig. 10(c) by building groups, and those in Fig. 10(d) by a single building. The change maps generated by CFMNet align most closely with the ground truth, accurately capturing the change patterns of different ground objects.

Synthesizing quantitative and qualitative results, HierFormer and CDMamba yield relatively low $F1$ among comparison methods, consistent with their suboptimal visual performance. This phenomenon stems from the characteristics of the SYSU-CD dataset: it encompasses not only single change types (e.g., individual buildings and roads) but also composite change types (e.g., building groups). The resulting scene complexity and diversity pose significant challenges for global context modeling. Although HierFormer adopts a convolutional–Transformer hybrid architecture, it introduces downsampling in self-attention computation to reduce compu-

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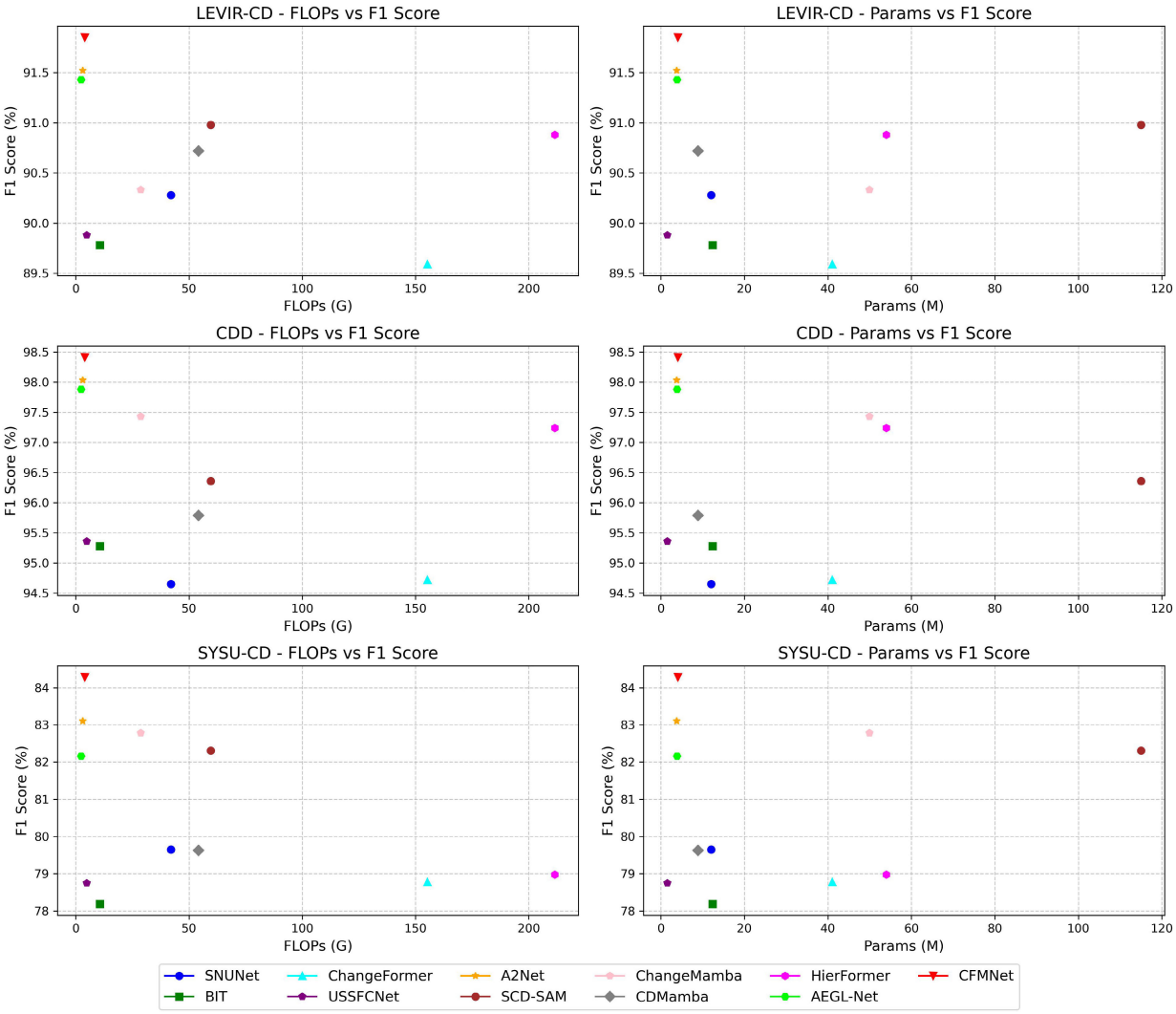


Fig. 11. Comparison of computational cost and parameter count among several CD methods on three datasets. Note: points closer to the top left indicate better performance (lower FLOPs/params and higher $F1$).

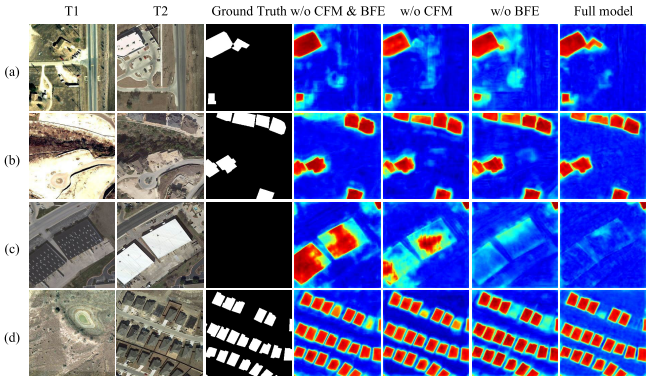


Fig. 12. Heatmap results of ablation experiments for the proposed innovations. (a)–(d) Present four sample sets selected from the LEVIR-CD test set, along with the corresponding heatmaps generated by models with different configurations.

tational costs, thereby compromising its global modeling capability. Likewise, CDMamba adopts a convolutional–Mamba hybrid architecture, yet its core reliance on traditional 1-D SSM inherently limits 2-D global context modeling. Consequently, both models inadequately capture semantic dependencies in composite scenarios, leading to overall inferior performance. CFMNet achieves superior results in both quantitative and qualitative analyses, attributable to the semantic-driven token interaction mechanism of the CFM block and the bidirectional feature streams design of the BFE module, which synergistically enhance the model's high-level semantic understanding and global context modeling.

4) Comparison of Computational Cost and Parameter Count: Fig. 11 presents the results of various methods on three datasets—LEVIR-CD, CDD, and SYSU-CD—focusing primarily on computational cost, parameter count, and $F1$. As can be observed from the figure, CFMNet achieves high $F1$ (91.85%, 98.41%, and 84.28%, respectively) while maintaining low computational complexity (4.05G FLOPs) and a small number of parameters (4.05M).

Among the compared methods, A2Net, AEGL-Net, and USSFCNet are categorized as lightweight approaches. USSFCNet possesses the fewest parameters at merely 1.52M. However, its accuracy (89.88%, 95.36%, and 78.75%) ranks

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TABLE IV
ABLATION STUDY ON THE PROPOSED INNOVATIONS

No.	Configure	Params (M)	FLOPs (G)	LEVIR-CD			SYSU-CD		
				F1 (%)	OA (%)	IoU(%)	F1 (%)	OA (%)	IoU(%)
#1	w/o BFE module & CFM Block	3.12	2.00	91.16	99.11	83.75	82.17	91.84	69.73
#2	w/o CFM Block	3.89	3.83	91.44	99.14	84.23	83.40	92.14	71.52
#3	w/o BFE module	3.28	2.23	91.42	99.13	84.20	83.40	92.34	71.52
#4	Full model	4.05	4.05	**91.85**	**99.18**	**84.93**	**84.28**	**92.70**	**72.82**

at a relatively low level across the three datasets. AEGL-Net and A2Net utilize MobileNetV2 as their backbones, achieving superior accuracy while maintaining low computational costs and parameter counts.

CFMNet employs KW MobileNetV2 as its backbone, enhancing the model's representation capability through dynamic KW convolutions while preserving computational efficiency. Furthermore, it incorporates the CFM module to model long-range dependencies with linear computational complexity, thereby avoiding the quadratic complexity overhead. Compared to A2Net and AEGL-Net, CFMNet achieves higher accuracy with only marginal increases in computational cost and parameter count. Specifically, on the SYSU-CD dataset characterized by composite scenarios, the $F1$ scores are improved by 1.18% and 2.12%, respectively. These findings demonstrate that CFMNet strikes a favorable balance between accuracy and efficiency, validating its effectiveness.

E. Ablation Studies

1) Ablation Study on the Proposed Innovations: In CFMNet, we propose two important innovations to improve the CD accuracy: the BFE module and the CFM block. To validate their effectiveness, we conduct ablation experiments on both the LEVIR-CD and SYSU-CD datasets. Four network variants are designed for comparison: Network #1 (the baseline) removes both novel components; Network #2 adds only the BFE module; Network #3 adds only the CFM block; and Network #4 integrates both, representing our CFMNet.

The results in Table IV validate the effectiveness of each innovation. Compared to the baseline (#1), introducing either the BFE module alone (#2) or the CFM block alone (#3) yields clear improvements, with $F1$ increasing by 0.28–0.26% and IoU by 0.48–0.45% on LEVIR-CD. This confirms that each component independently enhances the model's capability. More importantly, the full model (#4) achieves the best performance, surpassing the baseline by 0.69% in $F1$ and 1.18% in IoU. The fact that #4 outperforms both #2 and #3 indicates that the BFE module and CFM block provide complementary benefits. Furthermore, the consistent performance gains observed on the more challenging SYSU-CD dataset reinforce the robustness and generalization ability of the proposed innovations.

To further investigate the impact of the BFE module and the CFM block, Fig. 12 compares the model's attention heatmaps before and after their introduction. The results show that, compared with the baseline, incorporating either the BFE or CFM alone enhances the model's attention to actual change regions, although some attention remains on unchanged areas. When both components are combined, the resulting attention heatmaps achieve the closest correspondence to ground-truth change regions, exhibiting heightened responses within change areas while significantly suppressing activation elsewhere. These findings indicate that both the BFE module and the CFM block play crucial roles in CFMNet, and they exhibit a synergistic enhancing effect.

2) Ablation Study on the FM Module Design: To validate the internal design of the FM module, we conduct ablation studies to systematically evaluate the contributions of both the entire FM module and its core sorting operation. As shown in Table V, removing the entire FM module (#1) leads to a performance drop (e.g., −0.21% $F1$ on LEVIR-CD and −0.82% $F1$ on SYSU-CD). Disabling only the sorting operation within FM (#2) also results in a consistent decline across both datasets. These findings indicate that whether solely removing the sorting operation (while retaining the 1-D convolution) or completely eliminating the FM module, the model loses the capacity for interaction among semantically similar tokens, thereby impairing its discriminative ability. These results corroborate the critical value of the internal sorting operation and verify the rationality and effectiveness of the overall FM module design.

3) Ablation Study on the Subpaths in the BFE Module: We further analyze the internal structure of the BFE module by ablating its upstream and downstream subpaths. The results in Table VI indicate that removing the entire BFE module (#1) leads to a significant performance drop, with the $F1$ decreasing by 0.43% on LEVIR-CD (91.42% versus 91.85%) and by 0.88% on SYSU-CD (83.40% versus 84.28%). Independently disabling either the upstream (#2) or downstream (#3) subpath also causes clear degradation compared to the full model (#4). This indicates that both subpaths are indispensable to the module's functionality: the downstream subpath focuses on transmitting fine-grained details, while the upstream subpath reinforces high-level semantic information. Their fusion achieves effective multiscale feature enhancement, thereby validating the rationality and effectiveness of the BFE module's bidirectional feature stream design.

F. Comparative Evaluation of Lightweight Backbones

The encoder of CFMNet is based on KW MobileNetV2. To validate the effectiveness of KW MobileNetV2, we conducted comparative experiments with other lightweight backbones, including MobileNetV2 [43], EfficientNet [46], and ShuffleNetV2 [47]. The dataset used in the experiments is the LEVIR-CD dataset.

In the experiments, we replace the encoder in CFMNet with MobileNetV2, EfficientNet, and ShuffleNetV2, respectively.

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TABLE V
ABLATION STUDY ON THE FM MODULE DESIGN

No.	Configure	Params (M)	FLOPs (G)	LEVIR-CD			SYSU-CD		
				F1 (%)	OA (%)	IoU(%)	F1 (%)	OA (%)	IoU(%)
#1	w/o FM module	3.94	3.90	91.64	99.15	84.58	83.45	92.44	71.60
#2	w/o Sorting operation	4.05	4.05	91.69	99.16	84.65	83.52	92.64	71.71
#3	Full model	4.05	4.05	**91.85**	**99.18**	**84.93**	**84.28**	**92.70**	**72.82**

TABLE VI
ABLATION STUDY ON THE SUBPATHS IN THE BFE MODULE

No.	Configure	Params (M)	FLOPs (G)	LEVIR-CD			SYSU-CD		
				F1 (%)	OA (%)	IoU(%)	F1 (%)	OA (%)	IoU(%)
#1	w/o BFE module	3.28	2.23	91.42	99.13	84.20	83.40	92.34	71.52
#2	w/o Up stream	3.68	2.86	91.59	99.15	84.48	83.90	92.50	72.27
#3	w/o Down stream	3.68	3.45	91.64	99.15	84.57	84.04	92.56	72.48
#4	Full model	4.05	4.05	**91.85**	**99.18**	**84.93**	**84.28**	**92.70**	**72.82**

TABLE VII
COMPARISON OF KW MOBILENETV2 WITH OTHER LIGHTWEIGHT BACKBONES

No.	Backbone	Params (M)	FLOPs (G)	F1 (%)	OA (%)	IoU (%)
#1	MobileNetV2	4.58	4.82	91.63	99.15	84.55
#2	EfficientNet	6.38	5.06	91.65	99.16	84.59
#3	ShuffleNetV2	3.36	4.46	90.87	99.08	83.26
#4	KW MobileNetV2	4.05	4.05	**91.85**	**99.18**	**84.93**

TABLE VIII
COMPARISON OF CFM BLOCK WITH OTHER LONG-RANGE FEATURE EXTRACTION BLOCKS

No.	Block	Params (M)	FLOPs (G)	F1 (%)	OA (%)	IoU (%)
#1	w/o CFM Block	3.89	3.83	91.44	99.14	84.23
#2	ViT Block	3.99	6.20	91.57	99.15	84.46
#3	ViG Block	3.95	4.45	91.55	99.15	84.42
#4	VMamba Block	3.99	3.97	91.56	99.15	84.43
#5	ConvMamba Block	4.88	5.17	91.58	99.15	84.46
#6	CFM Block	4.05	4.05	**91.85**	**99.18**	**84.93**

The experimental results are shown in Table VII. It can be concluded that KW MobileNetV2 outperforms other lightweight backbones, with $F1$ that are 0.22%, 0.20%, and 0.98% higher than those of MobileNetV2, EfficientNet, and ShuffleNetV2, respectively.

G. Comparative Analysis of Long-Range Feature Extraction Methods

The CFM block proposed in this article effectively extracts long-range features through its semantics-driven similarity measurement. To objectively evaluate its performance, we compare it against several leading long-range feature extraction blocks, including the ViT block [17], ViG block [20], VMamba block [19], and ConvMamba block [15]. Specifically, we replace the CFM block in CFMNet with each of these blocks and conduct comprehensive comparisons.

As summarized in Table VIII, the CFM block achieves the best overall performance with the highest $F1$ (91.85%) and IoU (84.93%) among all compared methods. While the VMamba block exhibits marginally lower computational costs (3.97G versus 4.05G FLOPs), the CFM block demonstrates significantly better accuracy metrics. These results confirm that the CFM block provides a superior balance between accuracy and efficiency for CD tasks.

V. CONCLUSION

Based on the experimental results presented in this article, the proposed CFMNet offers an effective approach to remote sensing CD. The core component of this work, the CFM block, introduces a semantics-driven method for measuring token similarity, addressing limitations of distance-driven approaches while maintaining computational efficiency. In addition, CFMNet incorporates two other components: a BFE module for multiscale feature integration and dynamic KW convolution within the encoder for enhanced representation learning. Comparative experiments show that the $F1$ of CFMNet is at least 0.33%, 0.38%, and 1.18% higher than that of other advanced CD methods on three datasets: LEVIR-CD, CDD, and SYSU-CD, respectively. These results demonstrate the potential of CFMNet for practical CD applications.

We note that the precise localization of fine-grained targets (e.g., small objects and boundaries) remains a challenge, and our current design does not specifically optimize for it. Similarly, the issue of class imbalance inherent in CD datasets is not explicitly addressed. Addressing these aspects is important for building a comprehensively robust system. Therefore, our future work will focus on incorporating dedicated mechanisms to tackle these specific challenges, further advancing the robustness and practical utility of CD models.

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